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Invited paper

Impact of CEO media appearance on corporate performance in social media

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ABSTRACT

The impact of CEOs' media appearance in social media on corporate performance (financial performance) has received little research attention. In this paper, we propose the CEO media appearance indexes, namely CEO media coverage, CEO media transmission, and CEO media sentiment, and CEO searching attention indexes, namely media searching attention and user searching attention, then analyze the influence of CEO media appearance and CEO searching attention on corporate performance. The results show that the media transmission and media searching attention indexes have significant positive effects on corporate performance. The media sentiment and user searching attention indexes have significant negative effects on corporate performance; however, the same effect was not observed for media coverage. The effects of the above indexes were consistent with the interaction analyses. We discuss theoretical implications for research on CEO and corporate performance and management implications for corporate social media marketing.

1. Introduction

How we see a firm's top leader influences how we see their firm (Love, Lim, & Bednar, 2017). It is recognized that CEO decisions directly affect the success or failure of an enterprise (G. Jian & Fairhurst, 2016; Nohria & Khurana, 2010). CEOs are frequently featured in the business press, and some are even said to attain "celebrity" status through such exposure (Hayward, Rindova, & Pollock, 2004; Wade, Porac, Pollock, & Graffin, 2006). In short, corporate leaders can become the "face of the firm," appearing as the human force behind the firm's actions and outcomes (Strand, 2014).

Nowadays, CEOs are increasingly involved in social media activities. According to Weber Shandwick's survey (2012), CEO sociability increased from 36% to 66% between 2010 and 2012. CEOs' responsive and assertive communications on social media induce their followers to perceive the corporate leaders as amicable role models and caring friends, which in turn improves their trust of, satisfaction with, and advocacy for the company (Tsai & Men, 2017). The public is primarily motivated by reasons of thought leadership and task attraction when they engage with CEOs on social media (Men & Tsai, 2016).

A CEO's credibility and media appearance are significantly correlated with firm performance (Cianci & Kaplan, 2010; Francis, Huang, Rajgopal, & Zang, 2008; M. Jian & Lee, 2011; Tseng, Su, & Tsai, 2017). Moreover, the impact of CEO reputation is more persistent across different time periods and more comprehensive across different industries (Weng & Chen, 2016). Media attention on morally questionable CEO ethics negatively influence stakeholders' CSR motive attributions, which in turn results in increased cynicism that ultimately impacts CSR support intentions and behaviors (Ogunfowora, Stackhouse, & Oh, 2016). Negative news about

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a CEO is of critical importance to economic decisions, while positive news has a negligible impact (Scheidt, Henseler, Gelhard, & Strotzer, 2018). A CEO's terms and compensation are also affected by his/her media appearance (Conte, 2018). CEOs with high reputation are more likely to be dismissed than CEOs with low reputation when they perform poorly (Lee, 2007). As discussed above, in traditional media the influence of CEO reputation and firm performance has been recognized by many scholars, but the impact in social media has not been universally verified. In contrast with traditional media, the media coverage of CEOs in social media involves the relationship between CEOs and consumers, including transmissions and responses. Will these affect firms' economic performance?

In this paper we study the influence of CEO media appearance and searching attention index on corporate performance. We find the followings. First, CEO media appearance and searching attention index have significant impact on corporate performance. Specifically, media transmission and media searching attention indexes have significant positive effect on corporate performance; while media sentiment and user searching attention indexes have a significant negative effect. Second, CEO media appearance and searching indexes are consistently beneficial to firm performance. Our findings have interesting implications for corporate performance management. Firms should pay attention to users' transmissions and sentiment towards CEOs in social media and effectively manage the behavior related to CEOs.

The remaining sections of this paper are arranged as follows. The literature review is in Section 2. Section 3 introduces our research hypotheses. Research methodology is discussed in Section 4, and Section 5 discusses the analysis and results. Finally, we discuss our findings and contributions.

2. Literature review

The focus of this study is the influence of CEO media appearance in social media on corporate performance. We use two kinds of indexes to measure CEO media appearance: CEO Media Appearance in social media and CEO Searching Attention. We briefly introduce the literatures on CEO media appearance and CEO searching attention.

2.1. CEO media appearance

A few studies observed that CEO media appearance affects corporate performance, such as corporate financial performance, stock price, and corporate reputation. Table 1 contains a summary of the impact of CEOs on corporate or financial performance. CEO media appearance is measured by two types of index: CEO media coverage and tenor, which represents the media's recognition and emotional inclination towards a CEO; and CEO ability by industry-adjusted ROA, which represents a CEO's capability to create value for a firm in terms of financial performance. Previous studies were based on traditional media, such as newspapers and capital investment announcements.

The measurements of CEO media appearance based on news reports, such as tenor and total reports, did not take into account

Table 1
Summary of Papers on CEO Media Appearance and Corporate Performance.

Authors	Data	Media Variables	Conclusion
Weng & Chen, 2016	Taiwan's top 150 listed companies over the period 2003–2014	media coverage in six major newspapers in Taiwan industry-adjusted ROA (IAROA)	The impact of CEO reputation on the corporate financial performance is more persistent across different time periods and more comprehensive across different industries.
Love et al., 2017	372 large U.S. firms that appeared in Fortune magazine's annual "Most Admired Company" survey during the years 1991–1997	CEO media prominence; media tenor	Highly regarded CEOs enhance their firms' reputations, and CEOs who receive negative press coverage damage their firms' reputations; CEO prominence alone was not associated with higher firm reputation.
M. Jian & Lee, 2011	486 capital investment announcements from 1988 to 2007	CEO tenure; being appointed as CEO from outside of the firm; the number of business-related articles containing the CEO's name as returned by a search of the Factiva and Lexis-Nexis databases; industry-adjusted firm performance during the CEO's tenure	The stock market's responses to announcements of capital investments are more favorable for firms with more reputable CEOs. Moreover, CEO reputation mitigates the negative stock price reaction associated with announcements of capital investments by firms with high free cash flow and low growth opportunities.
Baixaulisoler & Sanchezmarin, 2014	534 Spanish listed companies in the period 2004–2009	rankings published annually by MERCO Monitor	Different scenarios depending on effectiveness of corporate governance. CEO reputation diffusion on TMT pay is strengthened or weakened by organizational governance effectiveness.
Francis et al., 2008	all S&P 500 companies over the 10-year period 1992–2001	Values of US News, IntlNews, press releases, and their sum, all articles	Using press coverage (media counts) to proxy for CEO reputation, found that more reputed CEOs are associated with poorer earnings quality.

Table 2
Summary of Papers on Influence of Searching Behavior.

Authors	Searching Index	Methodology	Conclusion
Zhi et al., 2011	LOG value of the user search index	correlation and time series analysis	User search index can be used to predict stock returns after 2 weeks.
Bordino et al., 2012	daily stock query data	correlation analysis time series analysisGranger causality test	The daily search index of stock code is related to the trading volume of the day.
Dimpfl & Jank, 2016	searching index in Google Trend	VAR model	There is strong linkage between stock market index and user search index; high search volume leads to high return price, and high return price leads to high search volume.
Bangwayo-Skeete & Skeete, 2015	daily user search index in the form of "hotels and lights"	seasonal autoregressive integral moving average model	User search index contains forecast information, especially the tourism industry.
Li, Pan, Law, & Huang, 2017	standardized search index for tourism keywords in Baidu and Google	generalized dynamic factor model (GDFM)	User searching index has predicting ability for tourism.

users' feedback and dissemination of reports. Media transmission can more directly depict the number of users receiving CEO reports. And user feedback is the direct expression of users' concerns and views, which may have a direct impact on other users. These factors should not be neglected in the study of the impact of CEO media appearance on firm economic performance. The emergence and application of social media makes it possible to obtain the above indicators. In this study, we conducted a specific study based on CEO media appearance in social media.

2.2. Searching attention

The research on searching attention has made quite remarkable findings in economics (Choi & Varian, 2012; Drake, Roulstone, & Thornock, 2012; Luo, Zhang, & Duan, 2012; Preis, Reith, & Stanley, 2010), disease prediction (Bordino et al., 2012; Ginsberg et al., 2009; Lazer, Kennedy, King, & Vespignani, 2014; Shaman & Karspeck, 2012), house pricing and sales (Wu & Brynjolfsson, 2015), unemployment (Askitas & Zimmermann, 2009; D'Amuri & Marcucci, 2010), and stock or tourism prediction (Yang, Pan, Evans, & Lv, 2015; Zhi, Engelberg, & Gao, 2011). The findings from these studies show that searching attention contains important information for forecasting economic environment, financial market, and enterprise performance (Table 2).

The above researches proved that searching attention on enterprise stocks or products has a certain predictive ability for firm economic performance. However, no one has studied the impact of CEO searching attention on firm performance. With the development of social media, CEOs increasingly connect and interact with users, which closes the distance between them. Under these circumstances, it is worth considering how the searching volume on CEOs affects firm economic performance.

3. Research hypotheses

The vast literature on leadership has proved that leaders are the driving force behind organizational behaviors and outcomes (Nohria & Khurana, 2010). Within managerial research, a central underlying idea is that the firm is a reflection of its leaders (Carpenter, 2004). Much work has shown that executive characteristics can impact a range of strategic choices and performance outcomes (Cannella, Finkelstein, & Hambrick, 2009). These general arguments about leaders also apply to CEOs. As firms' chief spokespersons, CEOs always provide interpretations of firm actions. The apparent power of CEOs is also featured in work on corporate saviors, founders, and charismatic CEOs, who are often perceived to have outsized influence on firm actions and outcomes (Maccoby, 2011). CEOs also gain prominence and perceived influence through the considerable attention they receive from the media, especially because journalists tend to present simplified storylines in which a firm's fortunes are attributed to the CEO and his/her leadership capabilities (Hayward, Rindova, & Pollock, 2010).

The media broadcasts are seen as critical in making information available about firms and their leaders (Bednar, 2012; Bednar, Boivie, & Prince, 2013). CEO media coverage affects ROA positively in the short run because of efficient operation, while negatively in the long run because of worse profitability (Akansu Cicon, Ferris, & Sun, 2017). Media coverage of CEOs can thus increase these leaders' prominence and affect perceptions of their quality. Leaders cannot affect corporate performance if evaluators know little or nothing about them. We predict that CEOs who receive more media coverage will generally confer more reputational benefits on their firms. Media coverage makes leaders more salient and closely identified with their firms, especially because at times it portrays them as powerful influences on firm outcomes and actions. As we have already described, leadership is positively valued in general (Meindl & Ehrlich, 1987). When CEOs become more salient, their firms should generally be seen in a more positive light. This argument draws strength from peoples' tendency to evaluate familiar objects more positively than unfamiliar ones. User familiarity impacts commenting and forwarding behaviors, so we believe that in social media, the amount of CEO media coverage is also an important factor affecting corporate performance.

H1: Companies whose CEOs have received more media attention will have stronger firm performance.

The media also acts as a social arbiter by making positive and negative judgments about the actors they cover, and these can serve as important signals regarding the quality of those actors by "rendering assessments of firms and individuals associated with them" (Wiesenfeld, Wurthmann, & Hambrick, 2008). On social media, the impact of CEOs on firms are becoming more and more important.

Compared with traditional media, CEO participation and reporting in social media have better advantages. On social media, the media account's coverage of business and business leaders can be quickly and effectively spread in a short time, and users can express their attitudes and views towards CEOs with feedback. Using social media data allows us to extract the number of media reports and identify the extent to which the media reports are concerned with the CEO, that is, the amount of CEO media coverage on the social media platform and the users' views. The number of tweets and retweets per day significantly affect both short-medium and long-term firm value, rather than firm performance (Zhou, 2017). Firm performance can be explained by the interaction effects of positive media tone and the change in CEO power, which is measured by the CEO pay slice (Huang, Roberts, & Tan, 2017). Media research has emphasized that positive press coverage can act as a valuable firm resource, in large part because it reduces the inherent uncertainty about firm and leader quality (Pollock & Rindova, 2003). We predict that when CEOs receive positive comments, they will confer reputational benefits on the firm, and conversely, they will inflict reputational damage when they receive negative comments. Thus, we propose the following two hypotheses.

H2: Companies whose CEOs receive more media transmissions will have stronger firm performance.

H3: Companies whose CEOs receive more positive media sentiment will have stronger firm performance.

The user searching index, as the agent of user's attention, contains the economic forecasting information. Scholars have verified its forecasting ability in specific fields, such as unemployment rate (D'Amuri & Marcucci, 2010), stock price (Ranco, Aleksovski, Caldarelli, Grčar, & Mozetič, 2015), and housing price (Wu & Brynjolfsson, 2015). Since user attention is limited, CEO searching to some extent represents the user's interest in a company. As the representative of the enterprise, CEO attention can be regarded as a kind of searching behavior on an enterprise, which represents a quantitative performance of the user's attention to the CEO and the enterprise. Moreover, social media coverage is more likely to provide users with relevant information on CEOs than traditional search engines/searching platforms. These search behaviors greatly affect users' decision-making and purchase behavior. Therefore, we believe that CEO-related searching indicators have an impact on corporate performance, leading to Hypothesis 4 and Hypothesis 5.

H4: Companies whose CEOs receive more user searching attention will have stronger firm performance.

H5: Companies whose CEOs receive more media searching attention will have stronger firm performance.

4. Methodology

4.1. Data collection and sample

To investigate the impact of CEO media coverage on corporate performance in social media, we obtained data from three different sources. First, we downloaded the CEO-related news reports and user comments from professional financial accounts (e.g. Caixin network; Fortune network; Business community and Finance and Economics) in Sina Microblog. Sina Microblog (www.sina.com) is the most widely used social media in China. Second, we got the user searching index from Baidu Index. According to the 40th report issued by the China Internet Network Information Center, 81.1% of Chinese internet users search information with Baidu. To a large extent, data from the Baidu Index represents the tendency of Chinese netizens' attention. Accordingly, we have reason to believe that data from Sina and Baidu are fully reliable in range and quality. Last, we obtained the firm financial performance from the CSMAR database, the largest and most accurate financial and economic database in China. It consists of eight series of stock, fund, bond, financial derivatives, listed companies, economy, industry, high-frequency data, and personalized data services.

As the second largest economy in the world, the changes of China's economic and financial markets have gradually attracted the attention of scholars. Liu and Qiang (2016) proved that media coverage can negatively moderate the impact of search volume on stock prices in China. Wang, Zhao and Kou (2017) found that abnormal search volume index predicts return reversal in the short term in China, where the effect is mainly caused by the searches of investors who use personal computers rather than mobile devices.

In our study, we chose the TOP200 corporates in China selected by Fortune Journal in 2018 as our research object. We chose the TOP200 firms for the following reasons. First, CEOs of large corporates receive more media attention. Second, users will know more about the company and CEO, and users will be more truthful and reliable when they publish comments on the CEO or the company in social media. We excluded corporates whose financial performance data was not available.

Finally, we got 1194 news about CEOs and 222,286 searching index numbers from 109 firms from January 1, 2011 to December 31, 2016 (Table 3).

4.2. Variables

4.2.1. CEO media appearance

CEO Media Coverage is a count of the CEO-related news during the focal year (Love et al., 2017). We selected the news about CEOs from professional financial newspapers in social media. Although CEO news has some repeatability in social media, we used the unique number of CEO-related news based on publishing time and publisher.

CEO Media Transmission is the number of retweets, comments, and "likes" (Zhou, 2017). These behaviors represent the user's concern for a CEO, which indicates the overall concerns caused by a CEO on social media.

We use principal component analysis to reduce the dimension of users' retweets, comments, and "likes." The results show that one principal component can be extracted from the three users' responses. The cumulative extraction load is 70.677%. We calculate the principal component of these three indicators as CEO media transmission.

CEO Media Sentiment We used the sentiment score of the users' comments to measure CEO media sentiment. We performed positive, negative, and neutral classification of user comments with a text emotional analysis method. Most positive comments

Table 3
Sample Industry Distribution.

Industry	Number of CEOs	Number of News	Number of Searching Index
Manufacturing	36	387	119,042
Mining	17	95	22,435
Finance	13	226	15,409
Wholesale and retail	11	98	19,714
Real estate	9	71	4463
Production and supply of electricity, water and gas	5	59	1578
Construction	5	25	34
Information transmission, computer service and software	5	105	34,730
Transportation, Logistics and Warehousing	4	44	1575
Agriculture Forestry Animal Husbandry Fisheries	2	19	1239
Culture, Sports and Entertainment	2	65	2067
Total	109	1194	222,286

Note. The industry classification is performed using Classification of National Economic Industries (GB/T4754) codes to identify 11 different industries.

express support and love for the CEO in the news. Negative comments express disapproval of a CEO's speech or decision or of the CEO himself. In neutral comments, users did not evaluate the CEO's views or decisions, they simply used @ to transfer or share the message. To analyze user comments, we use manual annotation and machine learning methods. We make up a training set by manual annotation, which accounts for 10% of the total number of comments. Then, the SVM classifier is realized by Python, and the user comment classifier is trained with the training set. The accuracy rate of the classifier by cross validation is 89.95%, which is acceptable. Finally, the remaining 90% of the user reviews are classified by the classifier. Sentimental value of the CEO-related news is obtained by the calculation of the positive comments.

4.2.2. CEO ability

IAROA We follow the approach proposed by Rajgopal, Shevlin, and Zamora (2006), who suggest that CEOs with better ability should enjoy better reputation. Rajgopal et al. use the industry-adjusted ROA (IAROA) to measure the ability of CEOs. The IAROA evaluates the performance of the CEO in the previous three years by the average difference between the ROA of the company for which the CEO works and the ROA of the industry in which the company is classified. If the IAROA is positive, it means that the CEO leads the company to perform relatively better than the whole industry, which is a signal that the CEO has better ability than other managers in the same industry. The detailed calculation is as follows:

$$IAROA_{i,t}=\frac{1}{3}\sum_{k=1}^{3}\left[ROA_{i,t-k}-\left(\frac{\sum_{j=1}^{N}ROA_{j,t-k}-ROA_{i,t-k}}{N-1}\right)\right]$$

where$IAROA_{i,t}$ is the industry-adjusted ROA for company i in the year t, $ROA_{i,t-k}$ is the ROA of company i in the last k year(s), and$\frac{\sum_{j=1}^{N}ROA_{j,t-k}-ROA_{i,t-k}}{N-1}$ is the average ROA of the industry excluding company i. N is the total number of companies in the industry.

4.2.3. Searching attention

First, we confirmed name combinations of CEO and Enterprise as searching keywords in order to eliminate the duplication of CEO names. Finally, we obtain two kinds of index: media searching index and user searching index (PC searching index and mobile searching index). To calculate the searching index, we adopt Zhi et al.'s method (2011). The formula is as follows:

Searching Attention=LOG(Baidu Searching Index)

4.2.4. Corporate performance

We calculate corporate performance through Tobin 's Q value. Tobin's Q is better at revealing the full picture of a firm's profitability by capturing the growth potential (Rao, Agarwal, & Dahlhoff, 2004). The calculation formula is as follows:

$$Tobin's\ Q=\frac{(MVE+PS+DEBT)}{TA}$$

where MVE is the value of the stock market in the company; PS represents the value of the preferred stock; DEBT is the net value of the company's liabilities; TA represents the value of the total assets of the company.

In addition, we examined the impact of CEO media appearance on corporate financial performance. In this paper we evaluated corporate financial performance through ROA, ROE, and EPS.

5. Control variables

We adopt the control scalar commonly used in the research of enterprise performance (Weng & Chen, 2016).

1. Firm size (LNSize): We use the total assets on the balance sheet to control for the firm's basic characteristics. The definition of size is:

LNSize=log(Total Assets)

2. Debit ratio (DR): We use the total assets on the balance sheet to control for the firm's basic characteristics. The definition of the debt ratio is:

$$\text{DR}=\frac{\text{Total Debt}}{\text{Total Asset}}$$

3.Market-to-book ratio (MTB): We use the total assets on the balance sheet to control for the firm's basic characteristics. The definition of the market-to-book ratio is:

$$\text{MTB}=\frac{\text{Total Market Value}}{\text{Total Equity Value}}$$

5.1. Model specification

The main purpose of this paper is to verify the impact of CEO media appearance and searching attention on corporate performance in social media. We used the panel data regression model to analyze the related impact mechanism.

$$Tobins\ Q_t = \alpha + \beta_1 CEO\ Media\ Appearance_t + \beta_1 CEO\ Ability_t + \beta_3 Searching\ Attention_t + \beta_4 Control\ Variables_t + \beta_5 Time\ Dummies + \beta_6 Industry\ Dummies + \varepsilon_{1t} \quad (1)$$

Similarly, we adopted the fixed-effect regression model for the influence of the above variables on the firm financial performance. The formula is as follows:

$$F_t = \alpha + \beta_1 CEO\ Media\ Appearance_t + \beta_2 CEO\ Ability_t + \beta_3 Searching\ Attention_t + \beta_4 Control\ Variable_t + \beta_5 Time\ Dummies + \beta_6 Industry\ Dummies + \varepsilon_{2t} \quad (2)$$

Where F indicate the corporate financial performance, we choose ROA, ROE and EPS.

We mentioned the two type of index: CEO Media Appearance and CEO Searching Attention, then we analyze the interaction effect of the above index. The formula is as follows:

$$Tobins\ Q_t = \alpha + \beta_1 CEO\ Media\ Appearance_t + \beta_2 CEO\ Media\ Appearance_t * D_{search} + \beta_3 Control\ Variables_t + \beta_5 Time\ Dummies + \beta_6 Industry\ Dummies + +\varepsilon_i \quad (3)$$

D_{search} is a dummy variable for high searching attention. If the searching attention is greater than the median value of searching attention in a specific year, $D_{\text{search}}$is equal to 1 in that year and 0 otherwise.CEO Media Appearance$_t * D_{\text{search}}$ thus captures the extra effect of CEO media appearance given the high searching attention. Similarly, consider the following model:

$$\text{Tobins}\ Q_t = \alpha + \beta_1 \text{CEO Searching Attention}_t + \beta_2 \text{CEO Searching Attention}_t * D_{\text{media}} + \beta_3 \text{Contron Variables}_t + \beta_5 \text{Time Dummies} + \beta_6 \text{Industry Dummies} + +\varepsilon_i \quad (4)$$

D_{media} is a dummy variable for CEO media appearance. If the CEO media coverage is greater than the median value of media coverage in a specific year, $D_{\text{media}}$is equal to 1 in that year and 0 otherwise.CEO Searching Attention$_t * D_{\text{media}}$thus captures the extra effect of CEO searching attention given a good CEO media appearance measurement.

6. Results

6.1. Descriptive Statistics and correlation analysis

In this study, we propose the CEO media appearance and searching indexes. CEO media appearance contains media coverage, media transmission, and media sentiment; the internet searching index contains media searching and user searching attention. Descriptive statistical analysis of these indicators is shown in Table 4.

Then we analyzed the correlation of key variables, as shown in Table 5. There is a significant correlation between CEO media appearance and Tobin's Q. The user searching index has a significant correlation with Tobin's Q. However, a relationship is not found in the media index. Because the user searching index was the sum of PC searching index and mobile searching index, the correlation between searching index is high; the correlation between user searching index and PC searching index is 0.923, and the correlation between user searching index and mobile searching index is 0. 965.

We also diagnose the multicollinearity of main variables. No significant collinearity was found. VIFs may provide additional insights to explain the high bivariate correlation. As a common rule of thumb, while a VIF value above 5 is a cause for concern, a value above 10 is regarded by practitioners to be a sign of serious or severe multicollinearity. In our study, the average VIF values for all models were below 5.

Table 4
Descriptive Statistics.

Variables	Mean	Std	Min.	Max.
CEO Media Appearance				
Media coverage	0.971	6.009	0	75
Media transmission	282.6073	2868.170	0	56613.0
Media sentiment	0.058	0.162	0	0.919
Searching Index				
Media index	2.754	2.595	0	9.836
User index	11.391	1.059	4.419	14.991
PC searching index	10.692	1.197	0	13.979
Mobile searching index	10.441	1.932	0	14.698
CEO Ability				
IAROA	0.015	0.062	−0.604	0.311
Corporate Performance				
Tobin's Q	0.242	0.363	−0.783	0.899
Financial Performance				
ROA	0.047	0.054	−0.140	0.351
ROE	0.119	0.122	−0.706	0.502
EPS	0.907	1.784	−4.828	17.819
Control Variables				
Firm Size	10.718	0.865	−0.140	0.351
DR	0.625	0.122	−0.706	0.502
MTB	0.002	1.784	−4.828	17.819

6.2. CEO media appearance and corporate performance

We chose the fixed-effects model based on a Hausman test that could reject the null hypothesis that the errors are not correlated with the explanatory variables (Prob > chi2 = 0.000).

Table 6 reports the regression results of equation (1), the relationship between CEO media appearance and corporate performance (Tobin's Q). Since we use two kinds of CEO media appearance measures, we add them to the regression individually as well as simultaneously. Notably, time and industry dummy variables were added to the equations, which means that the explanation of independent variables to dependent variables is the result of excluding the influence of time and industry factors. Clustering of standard error by industry is also considered in our model.

As reported, CEO media sentiment and media transmission have significant impact on corporate performance. CEO media sentiment has significant negative impact on corporate performance, while CEO media transmission has significant positive impact on corporate performance. However, the significant effect was not observed in CEO media coverage. The results indicate that in social media, CEO media coverage has no significant impact on a firm performance, but the user attention caused by news reports, that is, the media transmission and sentiment of the news, have a significant impact on firm performance. CEO media transmission represents the range of users who receive the information. These users include potential and target customers. Their dissemination of CEO-related reports represents concern for the firms, which will have an impact on the performance of the firm. CEO media sentiment is a kind of sentimental expression, which affects the purchase decision of the user and also has an impact on other users. Based on our calculation method, the range of media sentiment value is 0–1. Therefore 0.5 represents neutral feelings, < 0.5 is negative, and more than 0.5 is positive. Users' comments on CEO reports are more negative, which will have a negative impact on corporate reputation and performance. Thus, these results provide support to H2 and H3, but not to H1.

For searching index, the results reported that there is a significant correlation between CEO searching index and corporate performance. Media attention has a significant positive effect on firm performance. User searching attention has a significant negative impact on firm performance. For CEOs, media attention is generally related to their decisions on corporate policies, strategic development, and management events. These concerns can represent the initiatives of firms at the strategic and management levels, which will have a positive impact on corporate performance. Users' searching attention of CEOs will have a negative impact on corporates, mainly because users search more for specific products or firms themselves in the process of purchase decision-making. Searching for CEOs is more focused on news or events that are often negative. These results supported H5, but not H4.

By comparing the media coverage in social media and media attention, we identified that media attention has a positive impact on firm performance, but the impact of CEO media coverage on social media was replaced by users' transmissions and sentiments. Therefore, under the premise that social media is more and more widely used, when managing corporate performance through media coverage, CEOs should comprehensively consider the information access of different platforms.

Similarly, we validated the relationship between CEO media appearance and corporate financial performance, measured by return on assets (ROA), return on equity (ROE) and earnings per share (EPS). The results were shown in Tables 7–9. we also add all kinds of CEO media appearance and searching attention indexes to the regression individually as well as simultaneously. As reported in Table 7, CEO media appearance indexes have significant impact on return on assets (ROA); media and user searching indexes have the same effect. As shown in Table 8, CEO media appearance and searching indexes all have significant impact on return on equity (ROE). The same effect also validated on firm financial performance measured by earnings per share (EPS) in Table 9.

Table 5
Correlation Analysis of Key Variables.

Variables	1	2	3	4	5	6	7	8	9	10	11	12
1. Tobin's Q	1.000											
2. Media coverage	0.082 (0.035)	1.000										
3. Media transmission	0.046 (0.024)	0.833 (0.000)	1.000									
4. Media sentiment	0.161 (0.000)	0.304 (0.000)	0.185 (0.000)	1.000								
5. Media index	0.009 (0.821)	0.632 (0.000)	0.667 (0.000)	0.165 (0.000)	1.000							
6. User index	−0.071 (0.070)	0.463 (0.000)	0.489 (0.000)	0.094 (0.016)	0.616 (0.000)	1.000						
7. PC searching index	−0.061 (0.122)	0.343 (0.000)	0.312 (0.000)	0.096 (0.014)	0.506 (0.000)	0.923 (0.000)	1.000					
8. Mobile searching index	−0.072 (0.067)	0.505 (0.000)	0.567 (0.000)	0.085 (0.030)	0.637 (0.000)	0.965 (0.000)	0.791 (0.000)	1.000				
9. IAROA	−0.451 (0.000)	0.029 (0.494)	0.039 (0.361)	0.029 (0.500)	0.002 (0.961)	0.043 (0.321)	0.026 (0.547)	0.050 (0.243)	1.000			
10. ROA	−0.478 (0.000)	0.010 (0.810)	0.026 (0.506)	0.041 (0.296)	0.006 (0.869)	0.053 (0.248)	0.052 (0.187)	0.037 (0.344)	0.711 (0.000)	1.000		
11. ROE	−0.073 (0.064)	0.114 (0.003)	0.100 (0.010)	0.094 (0.016)	0.065 (0.098)	0.057 (0.144)	0.055 (0.165)	0.054 (0.167)	0.457 (0.000)	0.640 (0.000)	1.000	
12. EPS	−0.171 (0.000)	0.111 (0.005)	0.082 (0.035)	0.088 (0.025)	0.025 (0.525)	0.155 (0.000)	0.145 (0.000)	0.148 (0.000)	0.457 (0.000)	0.583 (0.000)	0.529 (0.000)	1.000

Note. The table presents pair-wise correlation coefficients among the key variables used in our empirical analysis. Significance levels (*p*-value) are displayed in parentheses.

Table 6
CEO Media Appearance and Corporate Performance.

Variables	Tobin's Q	Tobin's Q	Tobin's Q	Tobin's Q	Tobin's Q	Tobin's Q
CEO Media Appearance						
Media sentiment		−0.070***			−0.118***	−0.114***
		(−5.635)			(−2.309)	(−2.314)
Media coverage		−1.481 (−0.007)			0.0002 (0.786)	−0.0004 (−1.145)
Media transmission		0.012*** (9.296)			0.013*** (9.943)	0.014*** (10.757)
Searching Index						
Media searching index			0.011*** (2.900)	0.015*** (3.678)	0.015*** (3.399)	0.009*** (2.601)
User searching index				−0.033***	−0.035***	
				(−3.209)	(−3.299)	
PC searching index			−0.003***			−0.003***
			(−3.699)			(−4.075)
Mobile searching index			−0.009***			−0.009***
			(−1.621)			(−1.705)
CEO Ability						
IAROA	−0.516 ***	−0.515***	−0.499 ***	−0.456***	−0.459***	−0.498***
	(−3.120)	(−3.106)	(−3.035)	(−2.7560)	(−2.770)	(−3.022)
Control Variable						
FS	0.018 *** (4.221)	0.001 (0.358)	0.006*** (4.076)	0.007 *** (5.367)	0.005 *** (2.338)	0.003* (1.851)
DR	1.463*** (2.369)	1.460*** (2.365)	1.447*** (2.353)	1.445 (2.348) ***	1.441*** (2.339)	1.444*** (2.345)
MTB	−4.734 (−9.78)	−5.617 ***	−5.091***	−5.088 (−1.056)	−5.362***	−5.437***
		(−1.142)	(−1.059)		(−1.097)	(−11.143)
C	−0.716***	−0.659 (−4.502)	−0.617 ***	−0.385***	−0.341***	−0.576***
	(−5.253)		(−4.177)	(−2.264)	(−1.899)	(−3.614)
Time/Industry Dummy	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (Industry)	Yes	Yes	Yes	Yes	Yes	Yes
Observation	2616	2616	2616	2616	2616	2616
Adjusted R2	0.6986	0.7005	0.7046	0.7036	0.7045	0.7054

Note. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using fixed-effects estimation. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Table 7
CEO Media Appearance and Corporate Financial Performance Measured by ROA.

Variables	ROA	ROA	ROA	ROA	ROA	ROA
CEO Media Appearance						
Media sentiment		−0.007***			−0.007***	−0.006***
		(−3.008)			(−3.269)	(−2.804)
Media coverage		0.001 *** (2.512)			0.001*** (2.932)	0.0017*** (2.321)
Media transmission		0.049 *** (2.399)			0.051*** (2.408)	0.041 *** (2.007)
Searching Index						
Media searching index			0.003*** (3.901)	0.001*** (2.388)	0.003*** (4.066)	0.002*** (2.451)
User searching index			−0.008 ***		−0.008***	
			(−4.562)		(−4.717)	
PC searching index				−0.0001 (−1.601)		−0.097 (−0.829)
mobile searching index				−0.002***		−0.002***
				(−2.205)		(−2.053)
CEO Ability						
IAROA	0.407*** (1.473)	0.403***	0.420*** (1.531)	0.409*** (1.493)	0.416*** (1.528)	0.405*** (1.484)
		(1.465908)				
Control Variable						
FS	0.011*** (4.739)	0.012*** (4.952)	0.012*** (5.082)	0.011*** (4.809)	0.013*** (5.466)	0.013*** (5.137)
DR	−0.103***	−1.037***	−0.107***	−0.106 ***	−0.107***	−0.107***
	(−1.002)	(−1.013)	(−1.046)	(−1.037)	(−1.060)	(−1.044
MTB	6.869*** (8.503)	7.012*** (8.603)	6.878*** (8.596)	−0.106***	7.079*** (8.800)	7.053*** (8.723)
				(−1.037)		
C	−0.029***	−0.041***	0.047*** (1.679	−0.010 ***	0.033*** (11.220)	−0.028***
	(−1.270)	(−1.684)		(−4.115)		(−10.688)
Time/Industry Dummy	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (Industry)	Yes	Yes	Yes	Yes	Yes	Yes
Observation	2616	2616	2616	2616	2616	2616
Adjusted R2	0.6246	0.6307	0.6379	0.6316	0.6456	0.6370

Note. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using fixed-effects estimation. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Table 8
CEO Media Appearance and Corporate Financial Performance Measured by ROE.

Variables	ROE	ROE	ROE	ROE	ROE	ROE
CEO Media Appearance						
Media sentiment		−0.012*** (−2.800)			−0.014*** (−2.035)	−0.011*** (−2.625)
Media coverage		0.003*** (2.560)			0.003*** (2.718)	0.002*** (2.389)
Media transmission		0.123*** (1.984)			0.130 *** (2.000)	0.097*** (1.964)
Searching Index						
Media searching index			0.006*** (3.016)	0.005 *** (2.477)	0.006*** (2.689)	0.004*** (1.949)
User searching index			−0.012*** (−2.326)		−0.013*** (−2.55)	
PC searching index				0.006 *** (1.980)		0.006*** (1.975)
Mobile searching index				−0.009*** (−3.374)		−0.008 *** (−3.228)
CEO Ability						
IAROS	0.762 *** (9.179)	0.753 *** (9.113)	0.782*** (9.381)	0.761*** (9.276)	0.774*** (9.324)	0.751*** (9.167)
Control Variable						
FS	0.044*** (6.357)	0.043*** (57.807)	0.046*** (6.492)	0.046*** (6.583)	0.046*** (6.009)	0.0453 *** (6.143)
DR	−0.003 (−0.904)	−0.006* (−1.896)	−0.008 *** (−2.655274)	−0.011*** (−3.696)	−0.012 *** (−3.743)	−0.0138*** (−4.506)
MTB	18.8197*** (7.750)	18.37*** (7.495)	18.738 *** (77.084)	18.729*** (7.800)	18.560*** (7.572)	18.589*** (7.664)
C	−0.412*** (−6.023)	−0.393 *** (−5.385)	−0.299625*** (-3.491518)	−0.406*** (−5.50521)	−0.281*** (−3.120)	−0.401*** (−5.068)
Time /Industry Dummy	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (Industry)	Yes	Yes	Yes	Yes	Yes	Yes
Observation	2616	2616	2616	2616	2616	2616
Adjusted R2	0.3233	0.3343	0.3354	0.3414	0.3457	0.3488

Note. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using fixed-effects estimation. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Table 9
CEO Media Appearance and Corporate Financial Performance Measured by EPS.

Variables	EPS	EPS	EPS	EPS	EPS	EPS
CEO Media Appearance						
Media sentiment		−0.046*** (−4.553)			−0.057*** (−5.460)	−0.065*** (−6.345)
Media coverage		0.028*** (1.951)			0.0245*** (1.981)	0.024*** (1.989)
Media transmission		0.403*** (4.195)			0.196* (1.938)	0.314*** (3.254)
Searching Attention						
Media searching index			0.093*** (2.825)	0.088*** (3.08551)	0.089*** (2.646)	0.080 *** (2.704)
User searching index			−0.051*** (−6.269)		−0.070*** (−8.545)	
PC Searching index				0.001 (0.238)		0.002 (0.309)
Mobile searching index				−0.025*** (−6.168)		−0.027*** (−6.524)
CEO Ability						
IAROA	11.903*** (9.236)	11.729 *** (9.119)	12.050*** (9.305)	11.985*** (9.365)	11.896*** (9.192)	11.810*** (9.225)
Control Variable						
FS	0.521*** (4.796)	0.495*** (4.299)	0.509*** (4.665)	0.513*** (−2.246)	0.519*** (4.457)	0.518*** (4.497)
DR	−1.041*** (−2.165)	−1.069*** (−2.231)	−1.071*** (−2.221)	−1.076*** (−2.246)	−1.099*** (−2.282)	−1.096*** (−2.291)
MTB	46.026*** (12.206)	42.462*** (11.133)	38.352*** (10.161)	39.679 *** (10.599)	41.263*** (10.797)	41.968*** (11.072)
C	−4.308*** (−4.059)	−4.032*** (−3.554)	−3.824*** (−2.870)	−4.177*** (−3.631)	−3.693*** (−2.630)	−1.096*** (−2.291)
Time/Industry Dummy	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (Industry)	Yes	Yes	Yes	Yes	Yes	Yes
Observation	2616	2616	2616	2616	2616	2616
Adjusted R2	0.2472	0.2523	0.2607	0.2610	0.2624	0.2624

Note. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using fixed-effects estimation. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Table 10
Conditional Analysis of High and Low Searching Index.

Variable	Tobin's Q	ROA	ROE	EPS
Media sentiment	−0.127*** (−5.153)	0.061*** (14.919)	0.199*** (17.469)	1.375*** (8.342)
Media coverage	0.035 (1.331)	−0.002*** (−2.249)	−0.003* (−1.645)	−0.089** (−1.993)
Media transmission	0.014*** (2.256)	0.006*** (5.466)	0.014*** (4.725)	0.075* (1.763)
Media sentiment*Dmedia	−0.151*** (−5.964)	−0.056*** (−13.412)	−0.193*** (−16.549)	−1.009*** (−5.982)
Media coverage*Dmedia	−0.0343*** (−3.316)	0.003* (1.736)	0.005 (1.075)	0.124* (1.798)
Media transmission*Dmedia	0.018*** (2.704)	0.002** (2.1758)	0.013*** (4.254)	0.024 (0.566)
FS	0.055*** (40.879)	0.024*** (11.007)	0.041*** (6.573)	0.583*** (65.236)
DR	1.150*** (23.935)	−0.179*** (22.65)	−0.030*** (−13.545)	−2.043*** (−63.722)
MTB	−1.219*** (−2.591)	12.106*** (15.666)	23.275*** (10.720)	142.420*** (45.355)
C	−1.051*** (−7.446)	−0.126*** (−54.377)	−0.346*** (−53.128)	−4.409*** (−46.793)
Industry/Time dummy	Yes	Yes	Yes	Yes
Cluster (Industry)	Yes	Yes	Yes	Yes
Observation	2616	2616	2616	2616
Adjusted R2	0.5713	0.5719	0.1733	0.1144

Note. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using pooled regression estimation. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Therefore, it is determined that the proposed models on CEO media appearance and corporate performance have certain robustness.

6.3. Interaction analysis of CEO media appearance and firm performance

We examined the interaction between CEO media coverage and CEO searching attention in equations (2) and (3). In this study, we defined dummy variables for different indicators and analyzed the impact of CEO media coverage or CEO searching attention on corporate performance and financial performance under the interaction of dummy variables. Table 10 reports the results of equation (2) using media searching as the dummy variable in the interaction terms. The coefficients of media sentiment, media coverage, and media transmission present the impact of media appearance when media searching is poor. The coefficients of the interaction terms (media sentiment*Dmedia, media coverage*Dmedia, and media transmission*Dmedia) present the additional impact of media appearance when media searching is good. Surprisingly, we find that the coefficients of media sentiment and media transmission is significant, indicating that whether media searching index is poor or not, media sentiment and transmission have significant impact on firm performance. Besides, the coefficient of media coverage is significant when the media searching index is good. The results further confirm that regardless of the media searching index, when CEOs receive more media coverage (through transmissions), their companies perform better. Similarly, negative coverage of CEOs has negative impact on corporate performance.

Table 11 reports the results of equation (3) using CEO media coverage as the dummy variable in the interaction terms. The coefficients of CEO searching index present the impact of the searching index when media coverage is poor, and the coefficients of the interaction terms present the additional impact of the searching index when media coverage is good. We find that most of the searching index coefficients are significant, indicating that whether media coverage is poor or not, searching indexes have significant impact on firm performance.

We perform a pooled regression for equations (2) and (3). Regression results show that CEO media appearance and CEO searching attention have consistent significant impact on corporate performance and financial performance in different scenarios.

6.4. Robustness check

We test the robustness of our results by employing alternative specifications and considering various econometric concerns. Further, in order to address concerns about endogeneity and reverse causality, we report the results of running a panel vector autoregression (VAR) model.

6.4.1. Alternative specifications

We re-estimated our model with several alternative specifications. We estimate a random-effects model via maximum likelihood (RE-ML) and report the results in Table 12. The coefficient estimates from these specifications are qualitatively similar to our main results in Table 6, indicating the robustness of our results.

6.4.2. Reverse causality test

There is a reverse causal relationship between media coverage and corporate performance, that is, media are more willing to report on CEOs of corporates with high firm performance. In this study, we test the reverse causality with a panel vector autoregression model with exogenous variables (VARX). VARX has been used in previous studies (Chen, De, & Yu, 2015) and allows us to examine the dynamic interactions between CEO media coverage and firm performance. Chung, Animesh, Han, and Pinsonneault (2014) also validate the reverse causality problem with a panel vector autoregression model. The estimation results of our panel

Table 11
Conditional Analysis of High and Low Media Appearance.

Variable	Tobin's Q	Tobin's Q	ROA	ROA	ROE	ROE	EPS	EPS
Media searching index	0.013*** (28.445)	0.102*** (24.248)	0.001 (1.000)	0.012*** (6.264)	0.034* (1.979)	0.038* (2.030)	0.059* (2.219)	0.059* (1.982)
User searching index	−0.029*** (−25.818)		−0.012* (−2.0238)		−0.019* (−1.852)		0.004* (1.979)	
PC Searching index		−0.002*** (−2.286)		0.018** (5.314)		0.006 (1.181)		0.016 (0.321)
Mobile searching index		−0.009*** (−17.950)		−0.019* (3.145)		−0.007* (−1.889)		−0.009* (2.046)
Media searching index*Dmedia	0.015*** (13.126)	0.015*** (12.770)	−0.001 (−0.242)	−0.005** (−6.191)	−0.001* (−2.010)	−0.007* (−1.905)	−0.009 (−0.116)	−0.005* (2.061)
User searching index*Dmedia	−0.006 *** (−12.175)		−0.002 (−0.181)		0.002* (1.454)		0.002 (0.099)	
PC searching index *Dmedia		0.106 *** (4.845)		0.004* (2.156)		0.015*** (2.721)		0.069* (1.948)
Mobile searching index*Dmedia		−0.018*** (−7.921)		−0.004* (−2.123)		−0.014** (−2.591)		−0.070* (−1.865)
Observation	2616	2616	2616	2616	2616	2616	2616	2616
Adjusted R2	0.5729	0.5782	0.5582	0.5633	0.4593	0.4738	0.5136	0.5141

Notes. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using pooled regression estimation. Models include remaining set of control variables, time/industry dummies, and cluster (industry) dummies, but are not reported for brevity. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Table 12
Results of Alternative Specifications.

Variables	Tobin's Q	Tobin's Q	Tobin's Q	Tobin's Q	Tobin's Q	Tobin's Q
CEO Media Appearance						
Media sentiment		−0.014* (−1.122)			−0.012** (−2.101)	−0.023* (−2.034)
Media coverage		0.001* (1.302)			0.004 (0.19)	0.002 (0.234)
Media transmission		0.014** (2.112)			0.022** (2.177)	0.003*** (9.123)
Searching Index						
Media searching index			0.009** (2.480)	0.009** (2.556)	0.009** (2.311)	0.009** (2.267)
User searching index				−0.014** (−2.441)	−0.014*** (−2.467)	
PC searching index			−0.003* (−1.401)			−0.003* (−1.794)
Mobile searching index			−0.008* (−1.562)			−0.008** (−1.975)
CEO Ability						
IAROA	−0.458 *** (−2.746)	−0.459*** (−2.741)	−0.439 *** (−2.645)	−0.429*** (−2.593)	−0.4311*** (−2.592)	−0.439*** (−2.645)
Control Variable						
FS	0.003 (0.201)	0.001 (0.02)	0.002 (0.18)	0.001 (0.082)	0.001 (0.01)	0.002 (0.133)
DR	1.479*** (23.941)	1.477 (0.021)	1.464*** (23.74)	1.468*** (23.851)	1.465*** (23.75)	1.462*** (23.64)
MTB	−5.787 (−1.230)	−6.213 * (−1.311)	−6.076* (−1.392)	−6.226 (−1.345)	−6.359 (−1.355)	−6.148 (−1.300)
C	−0.691*** (−5.255)	−0.664*** (−4.750)	−0.586*** (−4.041)	−0.537*** (−3.667)	−0.5222*** (−3.345)	−0.576*** (−3.723)
Time/Industry Dummy	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (Industry)	Yes	Yes	Yes	Yes	Yes	Yes
Observation	2616	2616	2616	2616	2616	2616
Wald X^2	479.48	497.45	560.23	583.25	597.27	537.87

Notes. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using RE-ML model estimation. The sample is the 109 companies from TOP200 firms, which is selected by Fortune Journal in 2018, from 2011 to 2016. The unit of observation is firm–quarters.

Table 13
Panel VAR Coefficient Estimates.

	Dependent Variables in t					
Independent Variables in t-1	1	2	3	4	5	6
1. Tobin's Q_{t-1}	0.772*** (13.431)	0.107 (0.054)	0.013 (0.304)	−0.136 (−0.299)	0.497 (0.693)	−0.003 (−0.03)
2. Media $coverage_{t-1}$	0.006 (0.272)	0.562*** (3.421)	0.006 (−0.276)	−0.029 (−0.163)	0.183 (0.399)	−0.010 (−0.25)
3. Media $sentiment_{t-1}$	−0.009** (−2.379)	0.441 (0.142)	0.002 (0.047)	0.124 (0.296)	0.415 (0.463)	−0.066 (−0.96)
4. Media $transmission_{t-1}$	0.006*** (6.313)	−0.363 (−0.748)	−0.003 (−0.322)	−0.109 (−0.839)	−0.094 (−0.58)	−0.007 (−0.36)
5. Media searching $index_{t-1}$	0.009* (1.745)	0.082 (0.131)	0.005 (0.490)	0.061 (0.642)	0.638*** (3.347)	−0.024 (−1.28)
6. User searching $index_{t-1}$	−0.032*** (−8.442)	0.222 (0.109)	0.026 (0.795)	0.596* (1.792)	−0.348 (−0.57)	0.583***(7.776)

Note. Significance: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Models are reported using panel VAR estimation.

VARX model shown in Table 13 indicate that CEO media sentiment and media transmission have a significant effect on a firm's Tobin's Q after controlling for lagged endogenous variables and exogenous variables. Specifically, in the Tobin's Q equation, the coefficients estimate for media sentiment and media transmission are significant (-0.009, P < 0.05; 0.006, P < 0.05, respectively). In contrast, the coefficients for media coverage are not significant (0.006, P greater than 0.10). In addition, media searching index and user searching index also have a significant effect on firm performance (0.009, P < 0.10; −0.032, P < 0.01, respectively). Control variable and time/industry dummy variables are included in the estimation but are not reported for expositional brevity.

The panel VARX results are consistent with our main results and show that our findings are not driven by endogeneity, auto-correlations, or reverse causality.

7. Discussion and Conclusion

How CEOs influence their firm performance has received theoretical and empirical attention. However, prior research on the impact of CEO media appearance on corporate performance was based on traditional media; the impact in social media has not been systematically studied. In our study, we propose CEO media appearance and CEO searching attention indexes based on related research methods on information dissemination. Then we divided CEO media appearance into three categories: media coverage, media transmission, and media sentiment. CEO searching was divided into media attention and user attention according to the different subjects of concern. Based on CEO media appearance in Sina Microblog and the Baidu Index, we examined the impact of CEO media appearance in social media on firm performance. Our main findings are as follows.

First, CEO media transmission has a significant positive impact on corporate performance, whereas CEO media sentiment has a significant negative effect on firm performance. CEO media coverage has no impact. In social media, users receive a large amount of news and information, and they can express concerns or views through transmissions and comments. Media transmission represents the number of users who get and spread the information. Compared with media coverage, this indicator can better reflect the degree of user attention. Our results suggest that media coverage has no significant effects, but media transmission has a positive impact on enterprise performance. In addition, users' attitudes towards the news are crucial. The impact of user reviews on other users' behavior and product sales has been verified by many scholars. Our results prove that media sentiment has a negative impact on corporate performance. That is, in social media, CEO media coverage is no longer as important as in traditional media. Instead, media transmissions and sentiment affect firm performance significantly.

Second, the media searching index has a positive effect on corporate performance, while the user searching index has a negative effect. Searching indexes contain important information to predict economic changes and firm performance. The media pays more attention to CEOs' positive events or comments. The more media pays attention to a CEO, the more positive propaganda for the corporate and the CEO. However, users seldom pay attention to the CEO himself, they pay more attention to an corporate or products in their purchase decisions. Since more attention is paid to negative news, negative news on CEOs will have a negative impact on corporates.

Third, the influence of CEO media appearance and searching attention on corporate performance is consistent. The interaction analysis results show that the effect of CEO media appearance or CEO searching attention is similar in different scenarios. Although the above two indicators can better explain enterprise performance simultaneously in the model, the individual indicators also have a significant effect on enterprise performance. So, companies should consider both media appearance and searching behaviors in the process of managing CEOs in social media.

Implications for research There are some theoretical contributions in our study to the research on CEO media appearance and corporate performance. First, the findings from this study provide robust empirical support for the impact of CEO media appearance on firm economic performance in the social media context. Prior research on the impact of CEO media appearance on corporate performance is mainly based on traditional media. In traditional media, it is difficult to measure and evaluate the dissemination and user feedback. However, in social media, it is convenient to depict the actual amount of media dissemination and user comments. The research results also prove that media transmissions and sentiment are more important than CEO media coverage. Second, we proved that CEO media appearance and searching attention in social media have significant effects on corporate performance. Then, by comparing IAROA's interpretation of corporate performance (69.86% from Table 6) and CEO social media appearance on corporate performance (70.54%), we find that social media indicators can improve the interpretation of corporate performance. We can better explain and analyze corporate performance by analyzing the relevant indicators of CEOs in social media.

Management implications Our findings should be of practical value to managers and corporates alike who are seeking effective marketing channels in driving CEO media appearance and overall economic performance. We provide evidence to support the role of social media as an effective medium fulfilling this objective. With the development and application of social media, firms can enhance their performance by properly applying CEO media appearance. CEO media appearance includes media attention and users' transmissions and responses, which may include negative comments. Therefore, in the practical application process, appropriate measures should be taken.

Limitations In our study, we focus on the influence of CEO media appearance and CEO searching attention on corporate performance based on data from social media and online searching platforms. However, there are many types of social media platforms, as well as APPs initiated by different newspapers. In the follow-up study, the data of these platforms should be effectively integrated in order to understand the impact of CEO media exposure on corporate performance more comprehensively.

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